ELECTRONIC DEVICES AND CIRCUITS (EDC) · BJT THERMAL STABILITY PROJECT

Comparative Analysis of Thermal Runaway in BJT Circuits: Fixed Bias vs. Voltage Divider Bias

LTspice DC Temperature Sweep (.dc temp 25 150 1) validating emitter negative feedback and stability factors

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What thermal runaway is and why it matters

A self-reinforcing heating loop that fixed bias cannot break and voltage divider bias can.

What is thermal runaway?

A cumulative positive feedback loop where power dissipation increases junction temperature, boosting leakage current and gain, pushing the BJT into saturation or catastrophic physical destruction.

Project Objective

Demonstrate via LTspice simulation why Fixed Bias drifts rapidly into saturation while Voltage Divider Bias with R_E self-stabilizes the Q-point.

KEY PHYSICS DRIVERS

−2.0 mV/°C

V_BE temperature coefficient: falls as the junction heats

≈ 2× per 10°C

Reverse saturation current I_CBO roughly doubles

β rises

Current gain β increases with temperature

Open-loop versus negative-feedback biasing

Why one circuit drifts and the other corrects itself as the junction heats up.

OPEN LOOP

Fixed Bias

$$I_B = (V_{cc1} - V_{BE}) / R$$

As temperature rises, V_{BE} drops and β increases. With zero corrective feedback, I_C surges unchecked.

NEGATIVE FEEDBACK LOOP

Voltage Divider Bias

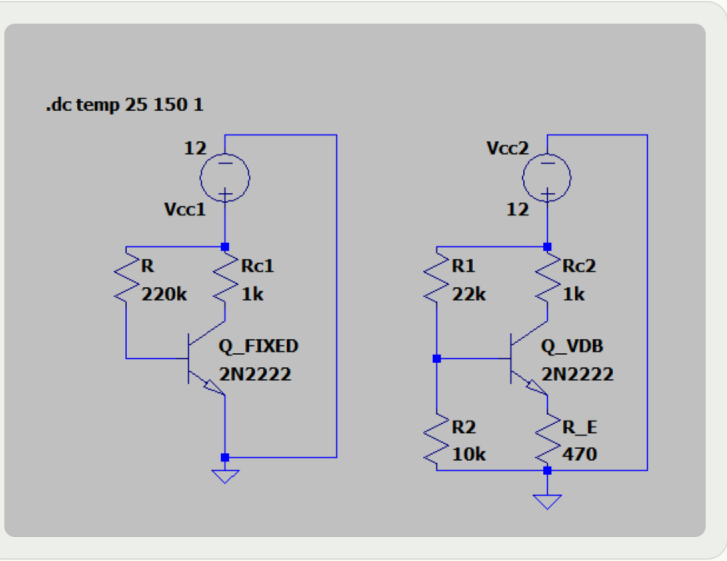
$T \uparrow \rightarrow I_C \uparrow \rightarrow I_E \uparrow \rightarrow V_E = I_E \cdot R_E \uparrow \rightarrow$

$V_{BE} = V_B - V_E \downarrow \rightarrow I_B \downarrow \rightarrow$ **I_C stabilized**

Highlight: R_E acts as an automatic internal feedback sensor that counteracts thermal expansion of collector current.

Dual-topology schematic and component roles

Both circuits share Vcc = 12 V, Rc = 1 kΩ and a 2N2222; only the bias network differs.



Vcc1 = Vcc2 = 12 V. Q_FIXED has its emitter direct to ground; Q_VDB reaches ground through R_E.

Component	Value	Topology	Functional role
R	220 kΩ	Fixed bias	Sets a fixed base current
Rc1	1 kΩ	Fixed bias	Collector load, sets V_CE1
R1	22 kΩ	Voltage divider	Upper divider arm, sets V_B
R2	10 kΩ	Voltage divider	Lower divider arm, sets V_B
Rc2	1 kΩ	Voltage divider	Collector load, sets V_CE2
R_E	470 Ω	Voltage divider	Emitter feedback, stabilizes I_C
2N2222	NPN BJT	Both circuits	Q_FIXED and Q_VDB, same model

Stability factor S for both topologies

A smaller S means the Q-point drifts less when leakage current changes.

$$S = \partial I_C / \partial I_{CO}$$

Definition: the change in collector current per unit change in the reverse leakage current I_{CO}.

FIXED BIAS

$$S = 1 + \beta$$

For $\beta \approx 200$, $S \approx 201$.

Extremely poor stability, open to runaway.

VOLTAGE DIVIDER BIAS

$$S = (1 + \beta)(1 + R_B/R_E) / (1 + \beta + R_B/R_E)$$

$R_B = R1 \parallel R2 = 22k \parallel 10k \approx 6.875\ k\Omega$

$R_B / R_E = 6875 / 470 \approx 14.63$

$S \approx (201 \times 15.63) / (201 + 14.63) \approx 14.5$

S falls from 201 to ≈ 14.5, about 13.8× more stable.

LTspice methodology and simulation bench

One schematic, two circuits and one temperature sweep for a direct cross-comparison.

Procedure

1. Built both topologies in one LTspice schematic for direct cross-comparison.
2. Applied the temperature sweep directive .dc temp 25 150 1.
3. Plotted collector currents I(Rc1) and I(Rc2) from 25°C to 150°C.

SPICE COMMAND

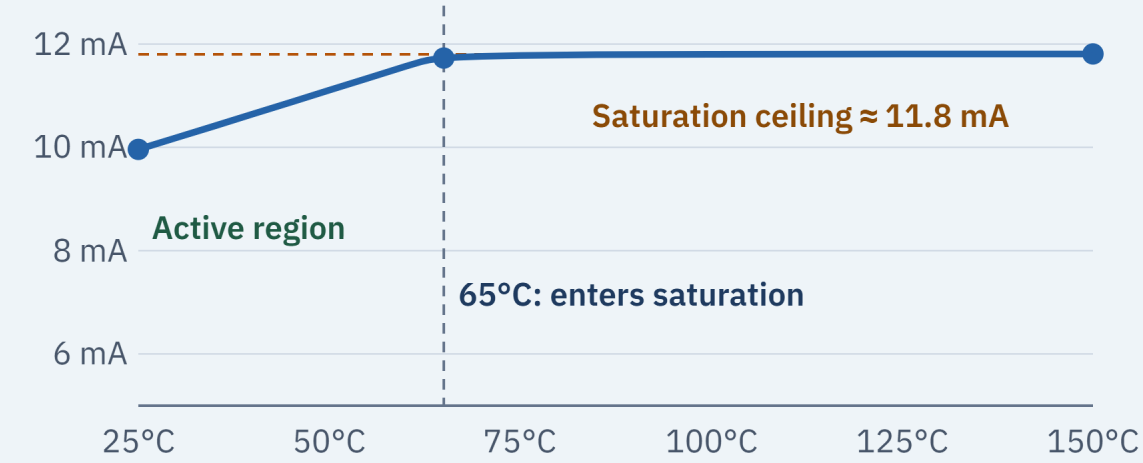
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.dc temp 25 150 1
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Start 25°C | Stop 150°C | Step 1°C | 126 points

Sweeping temperature natively exercises the temperature coefficients built into the Gummel-Poon BJT model (XTB, EG, IS, BF).

Fixed bias: I(Rc1) climbs into saturation

LTspice DC sweep, blue trace I(Rc1), 25°C to 150°C (Q_FIXED, 2N2222).



Temp	I(Rc1)	V_CE1	Region
25°C	≈ 10.0 mA	2.04 V	Active
45°C	≈ 10.9 mA	1.13 V	Active
65°C	≈ 11.7 mA	0.27 V	Sat. edge
85–150°C	≈ 11.8 mA	≈ 0.2 V	Saturated

Key insight

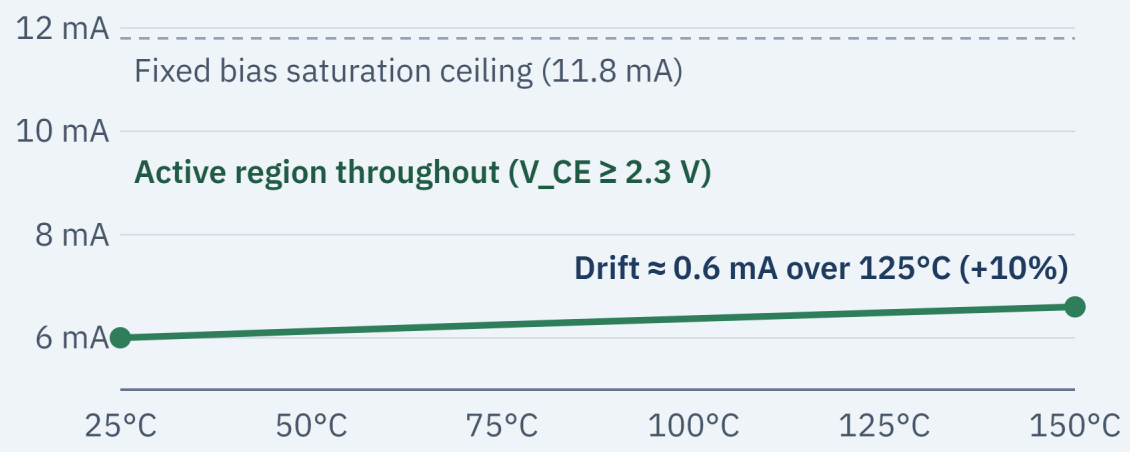
Above about 65°C fixed bias loses linear active operation. The ceiling is (12 V – 0.2 V) / 1 kΩ ≈ 11.8 mA.

07 · SIMULATION RESULTS: VOLTAGE DIVIDER BIAS (GREEN TRACE)

Voltage divider bias: I(Rc2) barely moves

LTspice DC sweep, green trace I(Rc2), 25°C to 150°C (Q_VDB, 2N2222).

I(Rc2) in mA versus temperature in °C (same scale as the fixed bias chart)



Temp	I(Rc2)	V_CE2	Region
25°C	≈ 6.0 mA	3.16 V	Active
65°C	≈ 6.2 mA	2.86 V	Active
105°C	≈ 6.4 mA	2.58 V	Active
150°C	≈ 6.6 mA	2.29 V	Active

Key insight

Across a 125°C heat rise, I(Rc2) drifts by only ≈ 0.6 mA (about 10%). The transistor stays in the linear amplifying region.

08 · DIRECT COMPARATIVE ANALYSIS

Side-by-side performance scorecard

Both circuits start in the active region; only one stays there across the full sweep.

Parameter / Metric	Fixed Bias: I(Rc1)	Voltage Divider: I(Rc2)	Verdict
Initial current at 25°C	~10.0 mA	~6.0 mA	Baseline: both active
Current at 150°C	~11.8 mA (saturated)	~6.6 mA (active)	VDB maintains linearity
Total current shift (ΔI_C)	+1.8 mA (+18%, hard clamp)	+0.6 mA (+10%, smooth)	VDB far superior
Q-point shift behavior	Drifts into saturation	Stays in the active zone	VDB immune to runaway
Stabilization mechanism	None (open loop)	Negative feedback via R_E (470 Ω)	VDB self-corrects

201 → 14.5

Stability factor S, fixed bias to voltage divider

1.8 vs 0.6 mA

Current drift (ΔI_C) over the full sweep

65°C vs none

Saturation onset: fixed bias vs voltage divider

Why voltage divider bias wins and what it costs

Stability comes at a price in AC gain, which a bypass capacitor recovers.

Why voltage divider bias is preferred

- Immune to temperature variations
- Tolerates β spread from batch to batch
- Survives transistor replacement without re-biasing

Tradeoff: the AC gain penalty

- R_E stabilizes DC bias
- It also feeds back on AC signals
- AC voltage gain drops as a result

PRACTICAL SOLUTION

Add an emitter bypass capacitor C_E (typically 10 μF to 100 μF) in parallel with R_E. It keeps the DC stability ($S \approx 14.5$) and shorts AC signals to ground for maximum AC gain.

10 · REAL-WORLD ENGINEERING APPLICATIONS

Where thermally stable biasing matters

Four areas where a drifting Q-point would cost performance or reliability.

Discrete audio preamplifiers

Keeps the Q-point centered and distortion-free while the stage warms up.

Automotive under-hood electronics

Sensor conditioning stages must work from −40°C to +125°C.

Linear power supply regulators

Prevents thermal runaway in BJT pass-transistor topologies.

RF and communication amplifiers

Stable bias networks in transceiver front-ends prevent thermal drift.

Links to Sustainable Development Goals

Reliable circuit design and simulation-based learning support two SDGs.

SDG 9

Industry, Innovation, and Infrastructure

Robust bias design prevents thermal overstress, reducing electronic device failure and mitigating hardware e-waste.

SDG 4

Quality Education

Interactive SPICE simulation brings core semiconductor physics to life, bridging textbook theory with engineering practice.

In this project: a full 25°C to 150°C stress test runs in software, so no real transistor is heated, damaged or wasted to learn the lesson.

12 · CONCLUSION & FUTURE SCOPE

What the simulation showed and what comes next

The LTspice sweep confirmed the textbook prediction across a 125°C rise.

Conclusions

- Fixed bias is thermally unstable and saturates rapidly as temperature rises.
- Voltage divider bias limits Q-point drift through negative emitter feedback.
- The .dc temp sweep is a rapid, accurate check of thermal stability.

Future scope

- Verify on hardware with a heating chamber and thermal imaging.
- Add an AC input and bypass capacitor C_E to study AC gain against DC stability.

Thank You